

# Transport Management System - Database Roadmap & ER Diagram

## 1. ER Diagram Summary: How Many Tables & Connections

Total Core Tables Built: 8 Tables.

How they connect to each other (Foreign Keys):

1. LrBilty (Lorry Receipt): The center table.
  - Connects to Customers (Consignor & Consignee).
  - Connects to Vehicles & Drivers.
2. EWayBill: Connects to LrBilty (1-to-1 connection).
3. Tracking: Connects to LrBilty (Many tracking updates for 1 LR) and Vehicles.
4. Customers: Connects to LrBilty.
5. Invoices\_Payments: Connects to LrBilty.
6. Vehicles: Connects to LrBilty & Tracking.
7. Drivers: Connects to LrBilty.
8. Users/Staff: Connects to LrBilty (Created by) & Tracking (Updated by).

## 2. Justification for TRACKING Table

Why is this table required instead of just using LrBilty status?

- Audit Trail: It keeps a complete timestamp history (e.g., Dispatched -> In Transit -> Checkpoint -> Delivered).
- Transparency: Customers and internal staff need clear logs of where the shipment is exactly.
- Exceptions: If a truck breaks down, the "Remarks" column in the tracking table allows you to log the delay reason.

Total Columns in Tracking Table: 7

| Column Name     | DataType     | Constraint   | Null Handling / Rule                                     |
|-----------------|--------------|--------------|----------------------------------------------------------|
| TrackingID      | INT          | NOT NULL PK  | Auto-increment primary key.                              |
| LrNumber        | VARCHAR(30)  | NOT NULL FK  | Connects to LrBilty.                                     |
| CurrentLocation | VARCHAR(200) | NOT NULL     | Exact city or checkpoint name.                           |
| Status          | VARCHAR(50)  | NOT NULL     | Created, In-Transit, Delayed, Delivered.                 |
| UpdateTime      | DATETIME     | NOT NULL     | Timestamp of the tracking log.                           |
| Remarks         | TEXT         | NULL ALLOWED | Empty normally. Fill data ONLY if exception (breakdown). |
| UpdatedByID     | INT          | NOT NULL FK  | Staff or Driver ID who updated this log.                 |

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## 3. Structure: LR\_BILTY Table (17 Columns)

This table stores the primary moving contract. Total Columns: 17

| Column Name       | DataType      | Constraint   | Null & Empty Row Logic                       |
|-------------------|---------------|--------------|----------------------------------------------|
| 1. LrNumber (PK)  | VARCHAR(30)   | NOT NULL     | Primary Key (e.g., LR-1001). Mandatory data. |
| 2. LrDate         | DATE          | NOT NULL     | Date receipt generated. Mandatory.           |
| 3. ConsignorID    | INT           | NOT NULL FK  | Sender Customer ID. Mandatory.               |
| 4. ConsigneeID    | INT           | NOT NULL FK  | Receiver Customer ID. Mandatory.             |
| 5. Origin         | VARCHAR(100)  | NOT NULL     | Start City. Mandatory.                       |
| 6. Destination    | VARCHAR(100)  | NOT NULL     | End City. Mandatory.                         |
| 7. MaterialDesc   | VARCHAR(255)  | NOT NULL     | Product details. Mandatory.                  |
| 8. TotalWeight    | DECIMAL(10,2) | NOT NULL     | Weight of goods. Mandatory.                  |
| 9. TotalPackages  | INT           | NOT NULL     | Number of boxes. Mandatory.                  |
| 10. FreightAmount | DECIMAL(10,2) | NOT NULL     | Total cost. Mandatory.                       |
| 11. AdvanceAmt    | DECIMAL(10,2) | NULL ALLOWED | EMPTY if no advance. Fill if paid.           |
| 12. BalanceAmt    | DECIMAL(10,2) | NULL ALLOWED | EMPTY if no balance. Fill if pending.        |
| 13. VehicleID     | INT           | NULL ALLOWED | EMPTY until a truck is assigned. Fill later. |
| 14. DriverID      | INT           | NULL ALLOWED | EMPTY until driver is assigned. Fill later.  |
| 15. EWayBillNo    | VARCHAR(20)   | NULL ALLOWED | EMPTY until govt ewaybill generates.         |
| 16. DeliveryDate  | DATETIME      | NULL ALLOWED | EMPTY while in transit. Fill UPON delivery.  |
| 17. Status        | VARCHAR(30)   | NOT NULL     | Created, Transit, Delivered. Mandatory.      |

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## 4. Structure: EWAY\_BILL Table (17 Columns)

Official government tracking mechanism. Total Columns: 17

| Column Name        | DataType      | Constraint   | Null & Empty Row Logic                        |
|--------------------|---------------|--------------|-----------------------------------------------|
| 1. EWayBillNo (PK) | VARCHAR(20)   | NOT NULL     | Govt generated number. Mandatory.             |
| 2. EWayBillDate    | DATETIME      | NOT NULL     | Generation Date. Mandatory.                   |
| 3. LrNumber        | VARCHAR(30)   | NOT NULL FK  | Linked LR. Mandatory.                         |
| 4. GenByGSTIN      | VARCHAR(15)   | NOT NULL     | Generator GST. Mandatory.                     |
| 5. SupplierGSTIN   | VARCHAR(15)   | NOT NULL     | Sender GST. Mandatory.                        |
| 6. RecipientGSTIN  | VARCHAR(15)   | NOT NULL     | Receiver GST. Mandatory.                      |
| 7. DispatchFrom    | VARCHAR(200)  | NOT NULL     | Start Address/PIN. Mandatory.                 |
| 8. ShipTo          | VARCHAR(200)  | NOT NULL     | End Address/PIN. Mandatory.                   |
| 9. Distance        | INT           | NOT NULL     | Distance in KM. Mandatory.                    |
| 10. TotalInvVal    | DECIMAL(10,2) | NOT NULL     | Invoice monetary value. Mandatory.            |
| 11. CustomDutyVal  | DECIMAL(10,2) | NULL ALLOWED | EMPTY if N/A. Fill data if cross-border.      |
| 12. CGSTAmount     | DECIMAL(10,2) | NULL ALLOWED | EMPTY/0 if exempt. Fill data if applicable.   |
| 13. SGSTAmount     | DECIMAL(10,2) | NULL ALLOWED | EMPTY/0 if exempt. Fill data if applicable.   |
| 14. IGSTAmount     | DECIMAL(10,2) | NULL ALLOWED | EMPTY/0 if exempt. Fill data if applicable.   |
| 15. VehicleNumber  | VARCHAR(20)   | NOT NULL     | Registered Truck Number. Mandatory.           |
| 16. TransporterID  | VARCHAR(50)   | NULL ALLOWED | EMPTY if self-transported. Fill if 3rd party. |
| 17. ValidUpto      | DATETIME      | NOT NULL     | Expiration of EWay Bill. Mandatory.           |

## 5. Summary of NULL Value Row Handling

Question: Which rows accept values, which rows want data, which will be empty?

Rule 1 (Future Data):

Columns like [VehicleID], [DriverID], [EWayBillNo], and [DeliveryDate] in LrBilty MUST BE EMPTY upfront. They only receive DATA when the action happens (e.g., when a truck arrives, data goes into VehicleID. When delivered, data goes into DeliveryDate).

Rule 2 (Conditional Data):

Columns like [AdvanceAmt], [CustomDutyVal], and [TransporterID] ONLY WANT DATA if the condition applies. If a customer paid no advance, the row MUST BE EMPTY (NULL) or zero. Do not use fake data.

Rule 3 (Mandatory Data):

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All NOT NULL rows (like Origin, Dest, Weight, LrNumber) MUST HAVE DATA immediately upon creation.